Fourier Transforms

DFT Examples

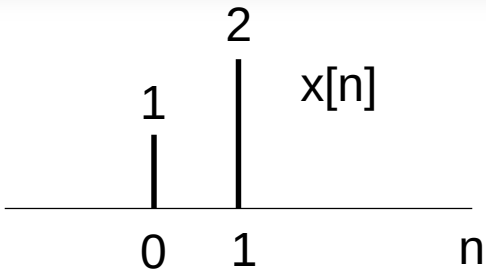
Ira A. Fulton Schools of Engineering
Arizona State University

Recall: DFT and IDFT

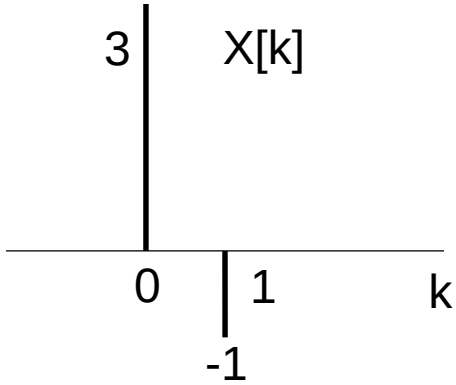
- $x[n] \xleftrightarrow{N-point\ DFT} X[k]$
- $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1$ **(DFT)**
- $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}, n = 0,1, \dots N-1$ **(IDFT)**
- **To avoid overlap in *x[n]*, *N* ≥ *L*.**

Example

- **Compute the 2-point DFT of *x[n]*:**



- **Answer:**

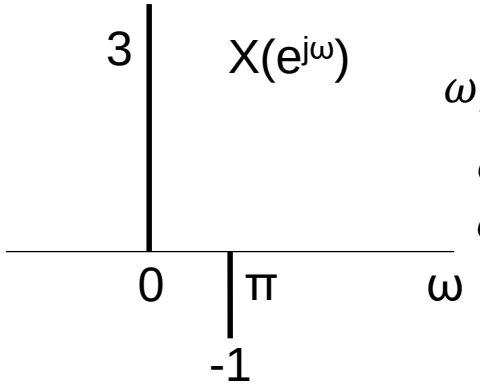


- $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \ldots, N-1$

- **For *N* = *2*:**

  - $X[0] = \sum_{n=0}^{1} x[n] e^{-j\frac{2\pi}{2}0n}$
    $= x[0]e^{-j\pi 0\cdot 0} + x[1]e^{-j\pi 0\cdot 1} = x[0] + x[1] = 3$

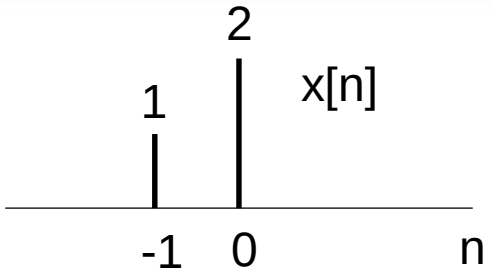
  - $X[1] = \sum_{n=0}^{1} x[n] e^{-j\frac{2\pi}{2}1n}$
    $= x[0]e^{-j\pi 1\cdot 0} + x[1]e^{-j\pi 1\cdot 1} = x[0] + x[1]e^{-j\pi}$
    $= x[0] - x[1] = -1$



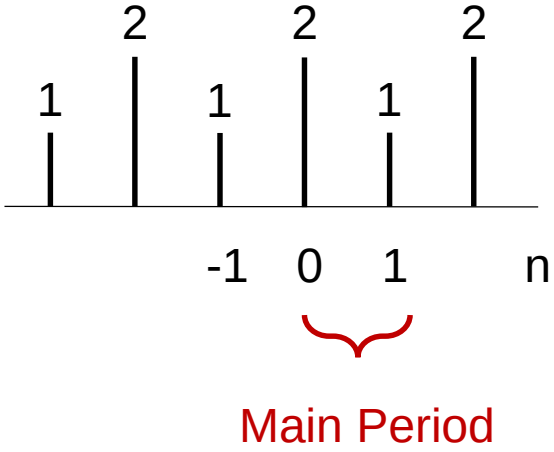
$$\omega_k = k\frac{2\pi}{N}$$
$$\omega_0 = 0$$
$$\omega_1 = \pi$$

Exercise

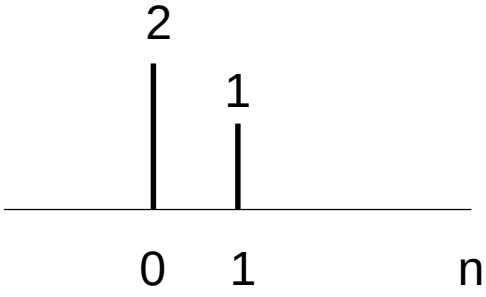
- Compute the 2-point DFT of *x[n]*:



- Since we have negative indices, we need to replicate *x[n]* with period *N = 2* (size of DFT), and take main period, *0 ≤ n ≤ N-1*.



- Now compute the 2−point DFT for:

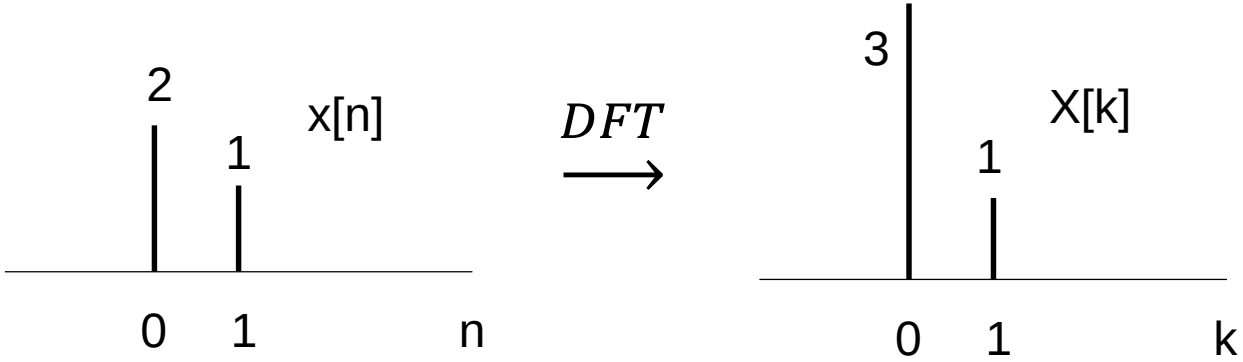


Answer

- $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1$

- **For *N = 2*:**

  - $X[0] = \sum_{n=0}^{1} x[n] e^{-j\frac{2\pi}{2}0n} = x[0]e^{-j\pi 0\cdot 0} + x[1]e^{-j\pi 0\cdot 1} = x[0] + x[1] = 3$
  - $X[1] = \sum_{n=0}^{1} x[n] e^{-j\frac{2\pi}{2}1n} = x[0]e^{-j\pi 1\cdot 0} + x[1]e^{-j\pi 1\cdot 1} = x[0] + x[1]e^{-j\pi}$
    $= x[0] - x[1] = 1$



Example

Compute the 4-point DFT of the following signal:

$$x[n] = \begin{cases} 2, n = 0 \\ 1, n = 1 \\ -1, n = 2 \\ 3, n = 3 \\ 0, otherwise \end{cases}$$

Answer

- $x[n] = [2, 1, -1, 3]$

- $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1$

- **For *N = 4*:**

  - $X[0] = \sum_{n=0}^{3} x[n] e^{-j\frac{2\pi}{4}0n} = x[0] + x[1] + x[2] + x[3] = 5$

  - $X[1] = \sum_{n=0}^{3} x[n] e^{-j\frac{2\pi}{4}1n}$

    $= x[0]e^{-j\frac{\pi}{2}1\cdot 0} + x[1]e^{-j\frac{\pi}{2}1\cdot 1} + x[2]e^{-j\frac{\pi}{2}1\cdot 2} + x[3]e^{-j\frac{\pi}{2}1\cdot 3}$

    $= x[0] + x[1](-j) + x[2](-1) + x[3](j) = 2 - j + 1 + 3j = 3 + 2j$

  - $X[2] = \sum_{n=0}^{3} x[n] e^{-j\frac{2\pi}{4}2n}$

    $= x[0]e^{-j\frac{\pi}{2}2\cdot 0} + x[1]e^{-j\frac{\pi}{2}2\cdot 1} + x[2]e^{-j\frac{\pi}{2}2\cdot 2} + x[3]e^{-j\frac{\pi}{2}2\cdot 3}$

    $= x[0] + x[1](-1) + x[2] + x[3](-1) = 2 - 1 - 1 - 3 = -3$

  - $X[3] = \sum_{n=0}^{3} x[n] e^{-j\frac{2\pi}{4}3n}$

    $= x[0]e^{-j\frac{\pi}{2}3\cdot 0} + x[1]e^{-j\frac{\pi}{2}3\cdot 1} + x[2]e^{-j\frac{\pi}{2}3\cdot 2} + x[3]e^{-j\frac{\pi}{2}3\cdot 3}$

    $= x[0] + x[1](j) + x[2](-1) + x[3](-j) = 2 + j + 1 - 3j = 3 - 2j$

Question

Which one of the following is NOT correct?

a) $e^{-j\pi/2} = -j$

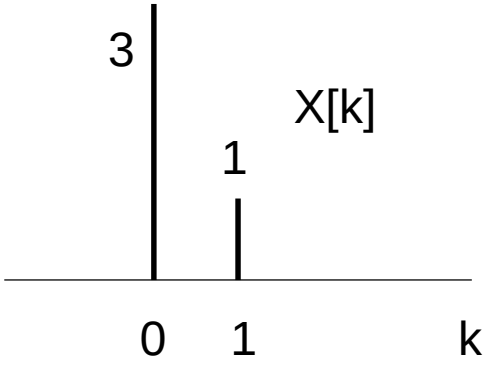
b) $e^{j\pi/2} = j$

c) $e^{-j\pi} = 1$

d) $e^{-j3\pi/2} = j$

Exercise

Compute the 2-point IDFT of *X[k]*:



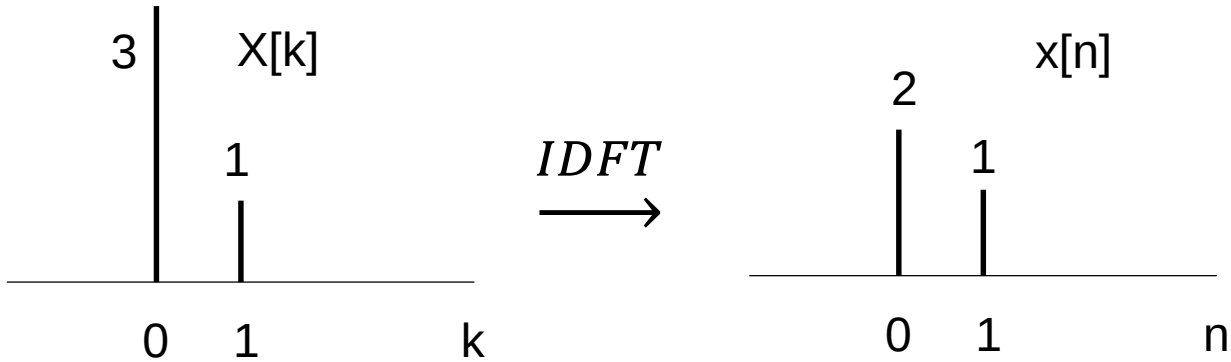
Answer

$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}, n = 0,1,\dots,N-1$

For *N* = *2*:

- $x[0] = \frac{1}{2}\sum_{k=0}^{1} X[k]e^{j\frac{2\pi}{2}k0} = \frac{1}{2}(X[0]e^{j\pi 0\cdot 0} + X[1]e^{j\pi 1\cdot 0}) = \frac{1}{2}(X[0] + X[1]) = 2$
- $x[1] = \frac{1}{2}\sum_{k=0}^{1} X[k]e^{j\frac{2\pi}{2}k1} = \frac{1}{2}\left(X[0]e^{j\pi 0\cdot 1} + X[1]e^{j\pi 1\cdot 1}\right) = \frac{1}{2}\left(X[0] + X[1]e^{j\pi}\right)$

$$= \frac{1}{2}(X[0] - X[1]) = 1$$



Ira A. Fulton Schools of Engineering

Arizona State University